

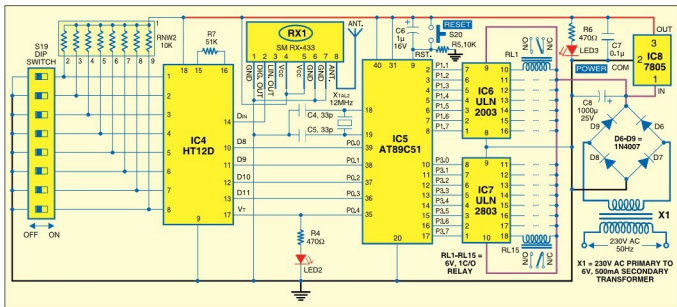


Fig. 3: Receiver circuit with SM RX-433 RF module (RX1)

HT12E and HT12D

HT12E and HT12D are CMOS ICs with a working voltage range of 2.4V to 12V. Encoder HT12E has eight address lines and four address/data lines. The

data set on these twelve lines (address and address/data lines) is serially transmitted when transmit-enable TE pin (pin 14) is low. The data output appears serially on D_{OUT} pin. Data is transmitted four times in succession.

The frequency of the pulses of data transmission may lie between 1.5 kHz and 7 kHz depending on the resistor value used between oscillator pins 15 and 16.

The internal oscillator frequency of decoder HT12D is 50 times the oscillator frequency of encoder HT12E. The values

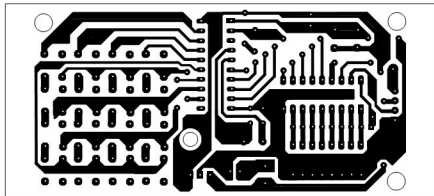


Fig. 4: An actual-size, single-side PCB for the transmitter circuit (Fig. 2)

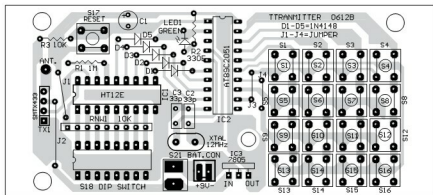


Fig. 5: Component layout for the PCB in Fig. 4

PARTS LIST

Semiconductors:

- | | |
|-----------|-----------------------------|
| K1 | - HT12E encoder |
| IC2 | - AT89C2051 microcontroller |
| IC4 | - HT12D decoder |
| IC5 | - AT89C51 microcontroller |
| IC6 | - ULN2003 relay driver |
| IC7 | - ULN2803 relay driver |
| IC3, IC8 | - 7805, 5V regulator |
| D1-D5 | - 1N4148 switching diode |
| D6-D9 | - 1N4007 rectifier diode |
| LED1-LED3 | - 5mm LED |

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 1.1-mega-ohm
- R2 - 330-ohm
- R3, R5 - 10-kilo-ohm
- R4, R6 - 470-ohm
- R7 - 51-kilo-ohm
- RNW1, RNW2 - 10-kilo-ohm network resistor

Capacitors:

- C1, C6 - 1 μ F, 16V electrolytic
- C2-C5 - 33pF ceramic disk
- C7 - 0.1 μ F ceramic disk
- C8 - 1000 μ F, 25V electrolytic

Miscellaneous:

- | | |
|---------------------------------------|--|
| X1 | - 230V AC primary to 6V,
500mA secondary
transformer |
| RL1-RL15 | - 6V, 1C/O relay |
| S1-S17, S20 | - Push-to-on tactile switch |
| S18, S19 | - 8-way DIP switch |
| S21 | - On/off switch |
| X _{TAL1} , X _{TAL2} | - 12MHz crystal |
| TX1 | - 433MHz RF transmitter
module (ASK) |
| RX1 | - 433MHz RF receiver module
(ASK) |